

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5333C

5 Credits :4

Program core

Source-grounded

IoT and Applications

Understand the role of IoT in various domains of Industry.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5333C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5333C.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automation and Robotics
Course code	5333C
Course title	IoT and Applications
Semester	5 Credits :4
Course category	Program core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

8 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Understand the role of IoT in various domains of Industry.

Course outcomes

CO	Official outcome	Study level
CO1	14 networking. Perceive ideas on the different connectivity Understanding	See official syllabus
CO2	14 technologies and Wireless Sensor Networks. Explain the basic concepts of cloud computing Understanding	See official syllabus
CO3	15 and fog computing Apply the basics of IoT for different applications Understanding	See official syllabus
CO4	15 like Smart Homes and Industrial IoT.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1
CO2	CO2 2 2
CO3	CO3 2 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the basic concepts of IoT and IoT networking.	2	As specified
Module 2	Sensor Networks.	2	As specified
Module 3	Explain the basic concepts of cloud computing and fog computing	2	As specified
Module 4	Industrial IoT.	2	As specified

MODULE 1

Explain the basic concepts of IoT and IoT networking.

This module is presented from the official syllabus scope for IoT and Applications. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the basic concepts of IoT and IoT networking.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▼ Define IoT 6 Understanding Enlist any three characteristics of IoT and

Define IoT 6 Understanding Enlist any three characteristics of IoT and is an official topic in Module 1 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define IoT 6 Understanding Enlist any three characteristics of IoT and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define iot 6 understanding enlist any three characteristics of iot and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Define IoT 6 Understanding Enlist any three characteristics of IoT and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Define IoT 6 Understanding Enlist any three characteristics of IoT and definition/meaning. principle, components or stages, steps, application appropriate. Technical terms English- Malayalam.

▼ 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming- Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless

8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies- Sensors-Characteristics of a Sensor Classification of Sensors-Actuators- Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless is an official topic in Module 1 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation.

Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding explain. characteristics of iot- applications of iot- iot categories- iot enablers and connectivity layers - baseline technologies-sensors-characteristics of a sensor classification of sensors- actuators-types of actuators-iot components and implementation. service oriented architecture-iot interdependencies-challenges for iot. iot networking connectivity terminologies, gateway prefix allotment-impact of mobility on addressing multihoming-deviations from regular web-iot identifications and data protocols. perceive ideas on the different connectivity technologies and wireless should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies- Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless definition/meaning . , steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Sensor Networks.

This module is presented from the official syllabus scope for IoT and Applications. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Sensor Networks.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▼ 8 Understanding HART.

8 Understanding HART. is an official topic in Module 2 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding HART. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding hart. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding HART. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-8 Understanding HART. ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം, steps, application ഫലപ്രസാദം ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English-ഫലപ്രസാദം ഫലപ്രസാദം.

▼ Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN-Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN.

Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN. is an official topic in Module 2 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-

NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN-Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the challenges in wsn 6 understanding introduction-ieee 802.15.4-zigbee-6lowpan-rfid-hart and wireless hart-nfc bluetooth-z-wave-isa 100.11a. wireless sensor networks introduction-components of a sensor node-modes of detection-challenges in wsn- sensor web-cooperation-behaviour of nodes in wsn-information theoretic self- management of wsn-social sensing in wsn-applications of wsn-wireless multimedia sensor networks. wireless nano sensor networks-under water acoustic sensor networks-wsn coverage. optimal geographical density control (ogdc) algorithm- stationary wsn-mobile wsn. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN-Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor

Aspect	What to prepare
	Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks- Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN. definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the basic concepts of cloud computing and fog computing

This module is presented from the official syllabus scope for IoT and Applications. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the basic concepts of cloud computing and fog computing
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▼ Explain public cloud. 8 nding Understa

Explain public cloud. 8 nding Understa is an official topic in Module 3 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain public cloud. 8 nding Understa using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain public cloud. 8 nding understa should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain public cloud. 8 nding Understa means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain public cloud. 8 nding Understa
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ **List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and**

List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and is an official topic in Module 3 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of

Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list out the advantages of fog. 7 nding introduction-architecture-characteristics-deployment models-public cloud-private cloudhybrid cloud-community cloud-multi cloud-distributed cloud-inter cloud-big data cloud-hpc cloud-service models-service management-cloud security. fog computing introduction-why fog computing-requirements of iot-architecture of fog-working of fog-advantages of fog-applications of fog-challenges in fog. apply the basics of iot for different applications like smart homes and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

MODULE 4

Industrial IoT.

This module is presented from the official syllabus scope for IoT and Applications. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Industrial IoT.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▼ Understand non-industrial applications of IoT 7

Understand non-industrial applications of IoT 7 is an official topic in Module 4 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand non-industrial applications of IoT 7 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand non-industrial applications of IoT 7 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand non-industrial applications of IoT 7 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Understand non-industrial applications of IoT 7 definition/meaning. steps, application. Technical terms English-.

▼ **Understand industrial applications of IoT 8 Smart Homes Introduction- Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT- Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book**

Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications

Understand industrial applications of IoT 8 Smart Homes Introduction- Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT- Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things:

Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited
 Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The
 R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4,
 Willy Publications is an official topic in Module 4 of IoT and Applications.
 Study the terminology first, then connect the stated purpose, sequence,
 components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT- Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand industrial applications of iot 8 smart homes introduction-origin of smart home-examples of smart home technologies-smart home implementation-home area networks-han elements-han standards-han architectures -han initiatives-smart home benefits and issues. industrial iot introduction-iiot requirements-design considerations-applications of iiot-manufacturing industry-health care service industry-transportation and logistics-mining-firefighting-smart dust-drones-futuristic farming-aerospace-energy networks- benefits of iiot-challenges of iiot. text / reference t/r book title/author honbo zhou, “the internet of things in the cloud: a middleware perspective”, t1 crc press,2012 t2 jeeva jose, “internet of things”, khanna book publishing co.(p) ltd, 1/e hanes david ,salgueiro gonzalo , grossetete patrick , barton rob , henry jerome, r1 “iot fundamentals: networking technologies, protocols and use cases for the internet of things”, pearson, 1/e. rajkamal, “internet of things: architecture and design principles”, mcgraw r2 hill (india) private limited daniel minoli, — “building the internet of things with ipv6 and mipv6: the r3 evolving world of m2m communications”, isbn: 978-1-118-47347-4, willy publications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements- Design Considerations-Applications of IIoT- Manufacturing Industry- Health Care Service Industry-Transportation and Logistics- Mining- Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A

Aspect	What to prepare
	Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Understand industrial applications of IoT
 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT- Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT- Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co. (P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet

of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications

definition/meaning

steps, application

Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Define IoT 6 Understanding Enlist any three characteristics of IoT and. (3 marks)**

► **Define or explain 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless. (3 marks)**

Module 2

► **Define or explain 8 Understanding HART.. (3 marks)**

► **Define or explain Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection- Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-**

WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN.. (3 marks)

Module 3

► **Define or explain Explain public cloud. 8 nding Understa. (3 marks)**

► **Define or explain List out the advantages of Fog. 7 nding Introduction-
Architecture-Characteristics-Deployment Models-Public Cloud-Private
CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-
Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service
Management-Cloud Security. Fog Computing Introduction-Why Fog
Computing-Requirements of IoT-Architecture of Fog-Working of Fog-
Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the
basics of IoT for different applications like Smart Homes and. (3 marks)**

Module 4

► **Define or explain Understand non-industrial applications of IoT 7. (3 marks)**

► **Define or explain Understand industrial applications of IoT 8 Smart
Homes Introduction-Origin of Smart Home-Examples of Smart Home
Technologies-Smart Home Implementation-Home Area Networks-HAN
Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart
Home Benefits and Issues. Industrial IoT Introduction-IIoT
Requirements-Design Considerations-Applications of IIoT-
Manufacturing Industry-Health Care Service Industry-Transportation and
Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-
Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text /
Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in
the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose,
“Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes
David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry
Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and**

Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define IoT 6 Understanding Enlist any three characteristics of IoT and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **8 Understanding HART..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain public cloud. 8 nding Understa.**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Understand non-industrial applications of IoT 7.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5333C. Verify institutional updates against the current official curriculum.